

# Expressions and Data Types

Introduction to computer programming

Management and Artificial Intelligence



# Why Python?

- A **high-level** language for **beginners**
- with a **clean syntax**, but...
- **Industrial strength** programming language used at thousands of companies (one of three official Google languages)
- Free, well documented, well supported

**Recall:** We will use Python 3 (there are non-backwards compatible differences with respect to Python 2). Further, Python 2 is now unsupported.

# Expressions and Statements

A (Python) program is composed by a sequence of:

- **definitions** (also **expressions**) which are **evaluated**
- **statements** (also **commands**) which are **executed**

In plain terms...

- **expressions** yield some results (e.g., `5 * 5`)
- **statements** instruct the interpreter to do something (e.g., print on screen, assign to variable)

Expressions and statements can either

- be directly written into a shell (**interactive mode**)
- be written into a text file and then loaded into the shell (**script mode**).

# Interactive vs. Script Mode

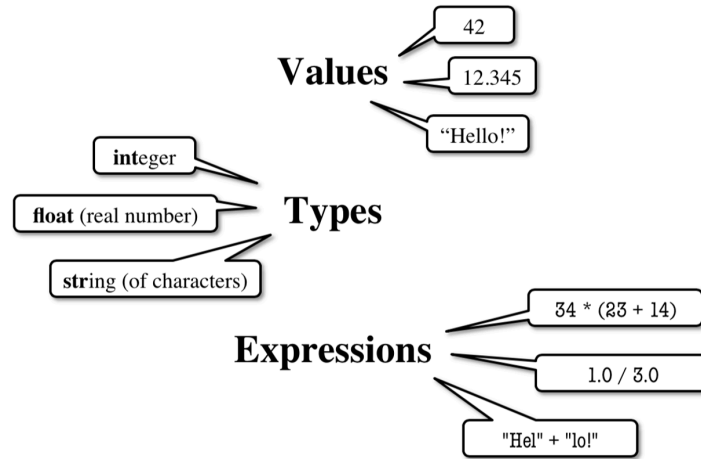
- **Interactive Mode:** perfectly fine for testing and learning.

Type your commands/expressions into the shell and Python will execute/evaluate those immediately:

|                                 |         |
|---------------------------------|---------|
| <code>&gt;&gt;&gt; 5 * 5</code> | ←input  |
| <code>25</code>                 | ←output |

- **Script Mode:** is what you normally do.
  - write the entire source code in a text file ending with the `.py` extension (say `myscript.py`)
  - `myscript.py` will be executed in its entirety by the Python interpreter

# Programming in Python



- programs manipulate **data objects**
- each data object has its own **type**, which defines the type of operations that can be done with them
- objects can either be:
  - **Scalar**: cannot be subdivided
  - **Non-scalar**: have an inner structure that can be accessed

# Scalar Objects

- `int` : represents integers (e.g., `5`)
- `float` : represents real numbers (e.g., `3.14`)
- `bool` : represents Boolean values (`True` or `False`)
- `NoneType` : special with only one value (`None`)

**Hint:** Use the `type()` function if you are not sure about the data type.

Examples:

---

```
>>> type(109) ←input
```

```
<class 'int'> ←output
```

---

```
>>> type(10.9) ←input
```

```
<class 'float'> ←output
```

# Type `int()`

Values:  $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

Integer literals look like a sequence of numbers with no periods nor commas.

Operations:

- `+`, `-`, `*`, `/`, `//` (floor division), `%` (modulus, remainder), `**` (exponentiation)
- `-` (unary minus, to negate a number)

Examples:

|                                |                                          |
|--------------------------------|------------------------------------------|
| <code>&gt;&gt;&gt; 7/5</code>  | ←input                                   |
| <code>1.4</code>               | ←output                                  |
| <code>&gt;&gt;&gt; 7//5</code> | ← <code>//</code> is the floor division! |
| <code>1</code>                 | ←output                                  |

Note In Python 3, operations between integers may not yield an integer.

# Type `float()`

Values: some (approximated) range of  $\mathbb{R}$

Note that:

- numbers with a "." are decimals (e.g., `2.0`)
- numbers with no decimal separator are integers (e.g., `2`)

Operations:

- `+`, `-`, `*`, `/`, `%` (modulus, remainder of division)
- `**` (exponentiation)
- `-` (unary, invert sign)

Operations between floats or between a float and an integer, always yield a float.

---

```
>>> 7.0/5.0 ←input
```

```
1.4 ←output
```

---

```
>>> 7.0//5 ← // is the floor division!
```

---

```
1.0
```

Python stores floating point numbers as base 2 (binary) fractions:

$$\{\text{number}\} = \{\text{integer mantissa}\} \cdot 2^{-\{\text{exponent}\}}$$

For example:  $0.125 = 1 \cdot 2^{-3} = 1 \cdot \frac{1}{2^3} = 1 \cdot \frac{1}{8}$ .

**Warning:** Not all numbers can be represented in base 2 fractions. Python chooses the best approximation it can. The same also holds for some numbers in base 10 fraction: e.g., what about 1/3?

Python may yield some **approximation errors**, that propagate as expressions go through.

Example:

|                                     |         |
|-------------------------------------|---------|
| >>> 0.1+0.2                         | ←input  |
| 0.30000000000000004                 | ←output |
| >>> 0.1+0.2+0.1+0.2                 | ←input  |
| 0.6000000000000001                  | ←output |
| >>> 0.1+0.2+0.1+0.2+0.1+0.2+0.1+0.2 | ←input  |
| 1.2000000000000002                  | ←output |

# Type `bool`

Values:

`True` and `False` (capitalized!)

Operations:

$$\text{not } a = \begin{cases} \text{True} & \text{if } a \text{ is False} \\ \text{False} & \text{if } a \text{ is True} \end{cases}$$

$$a \text{ and } b = \begin{cases} \text{True} & \text{if both } b \text{ and } a \text{ are True} \\ \text{False} & \text{otherwise} \end{cases}$$

$$a \text{ or } b = \begin{cases} \text{True} & \text{if } a \text{ is True or } b \text{ is True} \\ \text{False} & \text{otherwise} \end{cases}$$

Booleans are the output of comparisons between items (e.g., `int` or `float`):

- Order: `i < j`, `i > j`, `i <= j`, `i >= j`
- (In)equality: `i != j`, `i == j`

## Truth Table between two Boolean variables

| <b>a</b> | <b>b</b> | <b>not a</b> | <b>not b</b> | <b>a and b</b> | <b>a or b</b> |
|----------|----------|--------------|--------------|----------------|---------------|
| True     | True     | False        | False        | True           | True          |
| False    | False    | True         | True         | False          | False         |
| True     | False    | False        | True         | False          | True          |
| False    | True     | True         | False        | False          | True          |

# Type Conversions

Switching between `float` and `int` can be performed via **casting** (**explicit** conversion):

- `float(2)` converts value `2` (integer) to `2.0` (floating point)
- `int(2.9)` converts value `2.9` (floating point) to `2` (integer)

**Warning** `int()` does not round to the nearest integer: rather, it truncates the decimal part. To round to the nearest integer, use `round()` instead.

Python supports **widening** automatically (**implicit** conversion):

$$\underbrace{\text{bool} \rightarrow \text{int} \rightarrow \text{float}}_{\text{narrow to wide}}$$

- **Widening:** Python does it automatically, if needed. Example:

`1/0.5=2.0` because Python casts `1` to `float`

- **Narrowing:** Python never does it automatically. Because narrowing leads to information loss!

# Expressions

- **Expressions** are combinations of **objects** and **operators**
- an expression yields a **value**, which has its own **type**
- the simplest expression has the form:

`<object, operator, object>`

## Arithmetic Operations on `int` and `float`: a Recap

| Syntax            | Operator           | Output Type                                 |
|-------------------|--------------------|---------------------------------------------|
| <code>i+j</code>  | Sum                | if both of them are floats, result is float |
| <code>i-j</code>  | Difference         | if both of them are int, result is int      |
| <code>i*j</code>  | Product            | if either of them is float, result is float |
| <code>i/j</code>  | Division           | result is float (always)                    |
| <code>i%j</code>  | Remainder (Modulo) | if both of them are floats, result is float |
| <code>i**j</code> | Power of           | if both of them are int, result is int      |
| <code>i//j</code> | Floor division     | if either of them is float, result is float |

# Comparison Between `ints` and `floats`: a Recap

**NOTE:** Comparisons below evaluate to a `bool`.

| Expression             | Evaluation                                                          |
|------------------------|---------------------------------------------------------------------|
| <code>i &lt; j</code>  | → True if <code>i</code> is smaller than <code>j</code>             |
| <code>i &lt;= j</code> | → True if <code>i</code> is smaller than or equal to <code>j</code> |
| <code>i &gt; j</code>  | → True if <code>i</code> is greater than <code>j</code>             |
| <code>i &gt;= j</code> | → True if <code>i</code> is greater than or equal to <code>j</code> |
| <code>i == j</code>    | → True if <code>i</code> is equal to <code>j</code>                 |
| <code>i != j</code>    | → True if <code>i</code> is not equal to <code>j</code>             |

# Chained Comparisons

Comparisons can be chained arbitrarily and comparisons have the same priority:

$i < j < k$  is the same as  $i < j$  and  $j < k$

Since comparisons do not act in a pairwise fashion, 'weird' comparisons like

$i < j > k$

are perfectly valid.

# Operator Precedence

Precedence between operators can be 'forced' via parenthesis.

In principle, operators follow the **PEMDAS** order:

1. **P**arenthesis
2. **E**xponential
3. **M**ultiplication and **D**ivision
4. **A**ddition and **S**ubtraction

# The Big Picture

1. Parenthesis: `( ... )`
2. Exponentiation: `**`
3. Unary operators: `+` and `-`
4. Binary arithmetic: `*`, `/`, `//` and `%`
5. Binary arithmetic: `+` and `-`
6. Comparisons: `>`, `<`, `>=`, `<=`, `==`, `!=`
7. Logical not
8. Logical and
9. Logical or

**NOTE:** Same line means same precedence. Read ties left to right.

**HINT:** In case of emergency, use parenthesis!

